



Plane flies without moving parts

MIT engineers have designed a plane that can fly quietly without using any moving using ionic wind. <https://digit.in/dec18-72-1>



Laser can invite aliens

Existing laser and telescope technology on Earth can be used to attract aliens 20,000 light years away. <https://digit.in/dec18-72-2>

Good screen/ bad screen?

With screens surrounding the lives of children and adolescents increasingly, we need to understand how exactly they're affecting their lives

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ast month, we featured a special gadget in the Digit magazine - the IRL glasses. These were

glasses that can block out screens for you without blocking out anything else. While the mechanism behind it might be interesting, it brought us to wonder how significant the effect of screens around us is - especially on the younger members of our society. With smartphones, televisions and other forms of digital screens becoming ubiquitous and accessible to all age groups, children and adolescents are obviously no exceptions.

There's been a lot of attention on how increasing usage of gadgets with screens has been affecting various aspects of the lives of non-adults - including academic studies in a large number. By now, we should be pretty clear on whether screens are good or bad for the young ones, right? Turns out, it's more complicated than that.

SLEEP

How many times have you heard, that staring at smartphones before or while going to bed ruins your sleep cycle?

It's hard to avoid this piece of information especially when device and app manufacturers are introducing things such as Dark Mode, Reading Mode, Night Light etc. to reduce the amount of blue light that our eyes are subjected to in the fog end of the day. Things are no different when it comes to children, and the general consensus seems to be

relationship between the two, with other factors affecting sleep time relatively constant or on a downward curve.

However, there are studies that point in a totally different direction as well. A recently concluded study by the University of Oxford offers enough evidence contrary to the notion that screen time interferes with the regular sleep patterns of children. While the sample size of this study was smaller at around 50k, this sample was taken with the objective in consideration.



Effects of screen time on sleep might have been exaggerated

The author of the study, published in the Journal of Pediatrics, Professor Andrew Przybylski states that the previous studies have wrongly either used small sample sizes, or decided samples after obtaining the data, which could have led to misleading results. He also acknowledges that

that increased screen time is bad for sleep, more so for children.

There are studies that back this consensus. One conducted by researchers from San Diego State University and Iowa State University, involving the analysis of data from more than 350k adolescents, shows a 16-17% increase in likelihood of short-sleep duration over a period of time where screen time also increased significantly (digit.in/STStudy1). The study was able to draw a clear "exposure-response" re-

most of the data for such studies are obtained by consulting parents, which makes the data imperfect.

The Oxford study accommodates for the many of the factors by "adjusted for child-, caregiver-, household-, and community-level covariates". What this means is that the study accounted for things like the age, sex and ethnicity of the children, the education level and the financial situation of the immediate caregivers and the family involved, the availability and